

Open Project Space

# Program Syllabus

Fall 2026 - Spring 2027



**Lead Instructors:** Marvin Nguyen, Abishek Vinujudson

**Lab Supervisor:** Anika Saha

**Lab Instructors:** Anderson Kwon, Ashley Pock, Harsha Ganta, Hengjun Pei, Doan (Jackson) Nguyen, Radin Jafari, Shashank Sanigepalli, Steven Kuzhipala

## Course Description

Open Project Space (OPS) is a year-long technical course in embedded systems. We introduce engineering students to topics relevant to their coursework and potential career paths: integrated circuits, breadboarding, soldering, microcontrollers, embedded programming with C++, hardware communication, and PCB design.

Unlike a traditional college course, we aim to increase engagement with project-based learning. Students individually complete eight projects encapsulating the program's topics, culminating in one final capstone project where they design and build a custom rover. Regularly scheduled workshops and lectures complement the projects.

## Enrollment Requirements

Undergraduate students attending the University of California, Irvine, can apply for course enrollment. The deadline for application submissions is **October 2nd, 2026, at 11:59 PM**. Submitting an application does not guarantee acceptance to the program. Students who receive a course acceptance offer must pay the course deposit and fee before enrollment.

## Course Deposit + Fee

You will pay **\$120** (\$60 refundable deposit + \$60 course fee) at the beginning of the program. In exchange, you will receive a parts kit that you keep with the components needed to complete the course projects. Project deadlines are due the quarter they are assigned, and you will be refunded \$20 upon completing each quarter's required projects. By the end of the year, you can earn the full \$60 deposit back.

## Parts Kit Component List

- 9V Snap Connector (x1)
- Ball Caster Spacer Chassis Grip, 3D-Printed (x1)
- Battery Holder, 3D-Printed (x1)
- Battery, 9V (100 Count) (x2)
- BMP 280 (x1)
- Breadboard (x2)
- C (Male) to A (Female) Dongle (x1)
- Capacitor, 0.1uF (x2)
- Capacitor, 10uF (x2)
- Capacitor, 1uF (x2)
- DFPlayer A Mini MP3 Player (x1)
- DIP8 Socket (x1)
- ESP32 C3 (x2)
- Female-Male Jumper, Dupont, 20cm (x20) (x1)
- Gearbox Motor + Wheels (x2)
- Green LED, 2V (x3)
- HC-SR04 (x1)
- Header, 2.54mm, Female, 1x12 (x2)
- Header, 2.54mm, Female, 1x2 (x2)
- Header, 2.54mm, Female, Right Angle, 1x5 (x1)
- Joystick (x1)
- Jumper Wire Kit. 140pcs (x1)
- L293D (x1)
- LCD, 16x2 + I2C Backpack (x1)
- LED, RGB, Common Cathode (x1)
- Male-Male Jumper, Dupont, 20cm (x20)
- Metal Ball Caster (x1)
- Micro SD Adapter (x1)
- Micro SD, 1 GB (x1)
- Micro Servo (5V) (x1)
- NE555P (x2)
- PCB Perfboard, 4x6CM (x1)
- Photoresistor 5516, 5mm (x2)
- Piezo Buzzer, 1.5V (Passive) (x2)
- Potentiometer (x3)
- Reclosable Bags (x7)
- Red LED, 2V (x3)
- Resistor, 10K $\Omega$ , 5%, 1/4W (x5)
- Resistor, 1K $\Omega$ , 5%, 1/4W (x5)
- Resistor, 220 $\Omega$ , 5%, 1/4W (x5)
- Resistor, 390 $\Omega$ , 5%, 1/4W (x5)
- Resistor, 4.7K $\Omega$ , 5%, 1/4W (x5)
- Resistor, 470K $\Omega$ , 5%, 1/4W (x5)
- Sensor Mount, 3D-Printed (x1)
- Slotted Screwdriver (x2)
- SPDT Switch (x2)
- Speaker, 8 $\Omega$  0.5W (x1)
- Storage Box, 6qt (x1)
- Tactile Switches (x5)
- TM1637, 7 Segment Display, I2C (x1)
- USB-A to USB-C Cable (x1)
- Yellow LED, 2V (x1)

## Projects

Except for the Spring Quarter, projects are usually assigned every two weeks. These projects are typically due two weeks from their start date, while small assignments are due one week from their start date. Holidays may affect these times.

All the required projects must be completed before the final course deadline (TBD) for a student to be eligible for their deposit reimbursement.

Listed below are the projects assigned each academic quarter:

### Fall Quarter

| Project                            | Topics                                                                      | Description                                                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>LED There Be Light (Week 3)</b> | Circuits, Voltage, Current, Resistance, Ohm's Law, Breadboarding, Soldering | Build an LED circuit with a switch and solder it to a perfboard.                                                                    |
| <b>555 Piano (Week 5)</b>          | Circuit Analysis and Troubleshooting, Integrated Circuits, 555 Timer        | Create an electronic piano with the 555 Timer IC, and solder it to a printed circuit board.                                         |
| <b>RGB LED Wizard (Week 7)</b>     | Microcontrollers, ESP32, Arduino IDE, Pulse Width Modulation                | Build and program a dimmable RGB LED using the ESP32 and potentiometers.                                                            |
| <b>Arduino Radar (Week 9)</b>      | Microcontrollers (continued), ESP32, Servo Motors, Ultrasonic Sensors       | Build a radar system combining an ultrasonic sensor and servo motor, has two sweeping modes, and buzzes when an object is detected. |

### Winter Quarter

| Project                                  | Topics                                                        | Description                                                                                                                                      |
|------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>iPoduino v2 (Week 1)</b>              | Basic Computer Architecture, Arduino Libraries, UART Protocol | Build and program a music player with ESP32, customized to play your own tunes, now with a better speaker.                                       |
| <b>Wireless Weather Station (Week 3)</b> | Communication, I <sup>2</sup> C Protocol, SPI Protocol        | Build a wireless weather station that integrates an SPI sensor and wirelessly transmits data to an indoor display using a simplified WiFi setup. |

| Project                             | Topics                                           | Description                                                                                                                  |
|-------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Kitchen Timer (Week 6)</b>       | Interrupts, Timers                               | Build a kitchen timer using three buttons to start, reset, add, and subtract time, controlled through interrupts and timers. |
| <b>Capstone PCB Design (Week 8)</b> | KiCAD, Schematics, PCB Design, PCB Manufacturing | Design a PCB that you will use to control the Capstone Rover during Spring Quarter.                                          |

### Spring Quarter

| Project                                 | Topics            | Description                                                        |
|-----------------------------------------|-------------------|--------------------------------------------------------------------|
| <b>Capstone: RC Rover (All Quarter)</b> | Everything in OPS | Build and control a rover using both of your custom-designed PCBs. |

## Lectures

Lectures are hosted live every two weeks or pre-recorded unless otherwise stated. The exact dates, times, and locations are tentative. Each lecture discusses the topics relevant to upcoming projects and introduces the next project.

Listed below are the lectures hosted each academic quarter (subject to change):

### Fall Quarter

| Title                                                    | Date   |
|----------------------------------------------------------|--------|
| OPS Year-Long Roadmap & Introduction to Circuits         | Week 3 |
| Integrated Circuits & Debugging Equipment and Techniques | Week 5 |
| Microcontrollers & Introduction to ESP32                 | Week 7 |
| C/C++ Programming Review                                 | Week 9 |

### Winter Quarter

| Title                      | Date   |
|----------------------------|--------|
| Communication Protocols I  | Week 1 |
| Communication Protocols II | Week 3 |
| Interrupts and Timers      | Week 6 |
| PCB Design Concepts        | Week 8 |

### Spring Quarter

| Title                             | Date   |
|-----------------------------------|--------|
| Capstone Project: Getting Started | Week 1 |
| Other Microcontrollers            | Week 3 |
| Professional Development          | Week 5 |
| End of the Year Review/Reflection | Week 8 |

## Workshops

Workshops are hosted live or pre-recorded each quarter to supplement the projects. The exact dates, times, and locations are TBD. Each workshop serves as an interactive tutorial and a place to work with others.

Listed below are the workshops assigned each academic quarter (Dates/content are subject to change):

### Fall Quarter

| Title                              | Date   |
|------------------------------------|--------|
| Soldering/Desoldering I            | Week 4 |
| Soldering/Desoldering II           | Week 6 |
| Arduino IDE Setup & Potentiometers | Week 8 |

### Winter Quarter

| Title                 | Date   |
|-----------------------|--------|
| UART Workshop         | Week 2 |
| ESP32 Wifi Setup      | Week 4 |
| Winter Catch-Up       | Week 7 |
| PCB Design with KiCAD | Week 9 |

### Spring Quarter

| Title                   | Date   |
|-------------------------|--------|
| Capstone Rover Assembly | Week 2 |
| Capstone PCB Assembly   | Week 4 |
| Capstone Programming    | Week 6 |

# Agreement

By signing this document, you agree to the statement below:

*I hereby agree to pay the required sixty-dollar deposit and sixty-dollar course fee in exchange for the project materials and access to OPS lectures and workshops. I accept that a full reimbursement of the deposit is contingent on my completion of each project by the course's final deadline. I understand that my membership in Open Project Space is contingent on the completion of Project 1. I understand that I am eligible to refund my project kit only until the end of fall quarter if I choose not to continue with the program.*

*I understand that inappropriate behavior or any other form of misconduct inconsistent with the club constitution or bylaws may result in my removal from the OPS program without reimbursement of the deposit.*

Participant's Full Name (*Print*): \_\_\_\_\_

Participant's Signature: \_\_\_\_\_ / Date: \_\_\_\_\_

## Liability Release

*On this day of \_\_\_\_\_ intending to be legally bound hereby, the undersigned agrees and does hereby release from liability and to indemnify and hold harmless the Institute of Electrical and Electronics Engineers at the University of California, Irvine, and any of its officers as regards to the Open Project Space program and related workshop events. This release is for any and all liability for personal injuries, attorney fees and property losses or damage occasioned by, or in connection with any activity or accommodations for this program. The undersigned further agrees to abide by all the rules and policies promulgated by the Institute of Electrical and Electronics Engineers at the University of California, Irvine and/or its affiliate groups and vendors throughout the program.*

Participant's Full Name (*Print*): \_\_\_\_\_

Participant's Signature: \_\_\_\_\_ / Date: \_\_\_\_\_

If the participant is under 18 years of age, a parent or guardian must sign below:

Parent's Full Name (*Print*): \_\_\_\_\_

Parent's Signature: \_\_\_\_\_ / Date: \_\_\_\_\_